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Tyler Mccormack[1], Edward Beighley[2]

1. Northeastern University
2. Northeastern University

Abstract

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Applications, Limitations, and Insights for SWOT in Wave Dominated Nearshore Environments

Tyler McCormack[1], and Edward Beighley[1]

[1]Northeastern University, Boston, MA, USA

Corresponding author: Tyler McCormack (mccormack.ty@northeastern.edu)

Key Points:

- SWOT observes water surface elevation 500 m offshore with a root mean square error between 0.09 and 0.21 m
- SWOT KaRIN observes significant wave height with 0.19 and 0.18 m root mean square error 16 and 4 km offshore, respectively
- SWOT KaRIN significant wave height estimates remain valid until 1700 m from shore on a sandy alongshore uniform, wave dominated coast

Abstract

Water surface elevation (WSE) and significant wave height (SWH) data products from the Surface Water and Ocean Topography (SWOT) mission are evaluated against in-situ measurements in a nearshore environment for the period August 2023 to December 2025. Four SWOT WSE products (L3 LR SSH, L2 LR SSH, L2 HR Raster 100 m, and L2 HR Pixel Cloud) are compared to a NOAA tide gauge 500 m offshore (7 m water depth). The SWOT SWH data product is compared to four wave gauges in 7 to 26 m water depths, ranging from 600 m to 16 km offshore. Agreement between SWOT WSE and the tide gauge is best for the L3 LR SSH Expert product (RMSE = 0.09 m) and worst for the L2 HR Water Mask Raster product (RMSE = 0.21 m). SWOT SWH errors at wave buoys 4 and 16 km offshore are 0.19 and 0.18 m, respectively.

Plain Language Summary

The Surface Water and Ocean Topography (SWOT) satellite is a relatively new satellite that uses novel wide-swath altimetry to measure water surfaces over large areas compared to traditional nadir altimetry, that only measures directly below the satellite. SWOT has many exciting applications for measuring coastal processes due to large-swath technology, very fine measurement resolution, and the terrestrial and ocean products overlapping in the coastal zone. To better understand how SWOT can be used in coastal environments, we assessed the ability of the SWOT satellite to make accurate water level and wave height measurements close to shore by comparing SWOT and in-situ measurements. We found SWOT was capable of measuring water levels close to shore (500 m) with moderate accuracy, less than 10 cm error for the most refined product. SWOT demonstrated similar skill in measuring wave heights

farther offshore (4 and 16 km) with errors less than 20 cm. SWOT wave height measurements for locations less than 1,700 m offshore proved unreliable and were not used. Linear wave theory was used to translate valid SWOT wave height measurements closer to shore. This introduced more uncertainty producing errors between 25 and 30 cm.

1 Introduction

The Surface Water and Ocean Topography (SWOT) mission provides global measurements of water levels in rivers, lakes, and oceans (Fu et al., 2024). The Ka-band radar interferometer (KaRIn) evolves satellite altimetry from one-dimensional nadir measurements to wide-swath two-dimensional measurements of water surface elevations (Fu et al., 2024). The repeat orbit is approximately 21-days where the number of passes over a given waterbody within the 21-day orbit cycle varies by location (mostly due to latitude) (Birol et al., 2015). SWOT delivers two primary data products: (1) the flagship SWOT Low Rate (LR) products provide sea surface height (SSH), sea surface height anomaly (SSHA), and significant wave height (SWH) with a 2 km grid resolution; and (2) the terrestrial High Rate (HR) products provides lake and river WSEs with a 100 m grid resolution and anisotropic pixel cloud. The pixel cloud product has a footprint of 23 m in the along-track direction. The cross-track resolution varies from 60 to 10 m, decreasing in size as distance from nadir increases. These HR products are especially exciting for coastal environments where large geographic coverage is difficult.

SWOT launched Dec 16, 2022, and has shown it can resolve ocean characteristics at scales of 10-100's of kilometers (Morrow et al., 2019) such as SSH (Lichtman et al., 2025; Wang et al., 2025a; Wang et al., 2025b) and SWH with accuracy on the order of 10 cm (Ardhuin et al., 2025,

2024; Bohé et al., 2025; Ho et al., 2025; Peral and Chen, 2021; Turki et al., 2025). However, in the coastal region, wave evolution occurs at spatial scales ranging from 1-10 m (Dooley, 2026; Dooley et al., 2024; Raubenheimer et al., 1996; Birol et al., 2015), decreasing satellite wave height accuracy as measurements approach land (Ardhuin et al., 2024).

Considering the difference in scales between the open ocean and the coastal region, typical altimeter issues near the coast, and that SWOT was not specifically designed to resolve coastal processes, we work here to identify limitations in satellite measurements in the coastal zone. We compare SWOT WSE and SWH measurements with sensors deployed in water depths ranging from 7 to 26 m (600 m to 16 km offshore). We analyze the accuracy of SWOT data products in shallow water and discuss new coastal science opportunities.

2 Materials and Methods

2.1 SWOT Data Processing

We compared in-situ data collected at the Field Research Facility (FRF) in Duck, North Carolina to SWOT measurements from cycles 63 and 354, collected between August 2023 and December 2025 (Figure 1). We used four distinct SWOT products: (1) Level 3 Low Rate SSH Expert (AVISO/DUACS, 2024), (2) Level 2 Low Rate SSH Expert (Version C and D, SWOT 2024a, 2025a), (3) Level 2 High Rate Water Mask Raster 100 m (Versions C and D, SWOT 2024c, 2025c), and (4) Level 2 High Rate Water Mask Pixel Cloud (Versions C and D, SWOT, 2024b, 2025b). Each of the four products were used for the WSE analysis. Ony the Level 2 Low Rate SSH Expert product delivers SWH measurements, accordingly, SWH from this product was compared at four different locations. Each product was processed separately to account for differences in resolution,

variables, corrections natively applied, and quality flags. The four products share similar processing steps, with the specific nuances highlighted in the product specific subsections. All products were filtered to include only data in a 600 km^2 area adjacent to the FRF that met the quality flag criteria. The filtered data were converted to FRF coordinates, a local coordinate system extending offshore (positive x) and alongshore north (positive y).

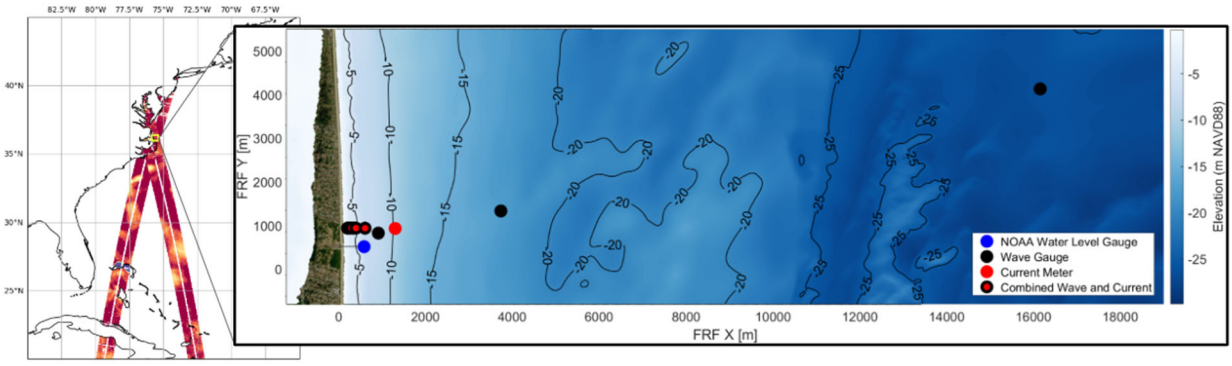


Figure 1. Map of the East Coast of the United States with SWOT passes 63 and 354 showing L3 LR SSH Unsmoothed sea surface height anomaly (SSHA) with the Field Research Facility (FRF) called out on the right showing the research pier (solid black line), NOAA tide gauge (blue circle), and y= 940 m transect of wave gauges (black circles). The color bar shows water depth (NAVD88) with 5 m depth contours

2.1.1 L3 LR SSH Expert 2 km

The Level 3 LR product is a more refined (compared to L2 products), multi-sensor merged product that includes a machine learning based denoising filter. SWOT data is flagged as good, suspect, degraded, or bad (JPL, 2025). The SSH from this product was filtered to exclude any data that were not considered "Good" by the quality flags. The WSE relative to NAVD88 was calculated using equation 1:

$$WSE_{NAVD88} = SSHA_{filtered} + MSS + DAC + Tide_{Ocean} - Geoid + EGM2NAVD88 \quad (1)$$

where WSE_{NAVD88} is the total water level relative to the NAVD88 datum, $SSHA_{filtered}$ is the sea surface height anomaly with all corrections, calibrations, data selection, and noise reduction

(Unet model) applied, MSS is the mean sea surface above the reference ellipsoid, DAC is the dynamic atmospheric correction, $Tide_{Ocean}$ is the geocentric ocean tide height (FES 2022), Geoid is the height of the geoid above the reference ellipsoid, and EGM2NAVD88 is a local conversion from EGM2008 datum to NAVD88 (0.24 m in this case).

2.1.2 L2 LR SSH Expert 2 km

For the L2 Low Rate SSH Expert product, SSH and SWH data classified as degraded or worse were excluded from WSE calculations. The water level relative to NAVD88 was calculated using equation 2:

$$WSE_{NAVD88} = SSH - Geoid + Xcor + SSB + SET + LT + PT + EGM2NAVD88 \quad (2)$$

where SSH is the sea surface height above the reference ellipsoid, Geoid is the geoid height above the reference ellipsoid as reported by this product, *Xcor* is the height correction from the crossover calibration, SSB is the sea state bias correction, SET is the solid earth tide height, LT is the load tide, PT is the pole tide, and $EGM2NAVD88$ is a local conversion from EGM2008 datum to NAVD88. The SWH product was used as is after filtering.

2.1.3 L2 HR Water Mask Raster 100 m

For the L2 HR Water Mask 100 m Raster product, data classified as bad were excluded. The water level relative to NAVD88 was calculated using equation 3:

$$WSE_{NAVD88} = WSE_{EGM2008} + EGM2NAVD88 \quad (3)$$

where $WSE_{EGM2008}$ is the water surface elevation relative to EGM2008 and *EGM2NAVD88* is a local conversion from EGM2008 datum to NAVD88.

2.1.4 L2 HR Water Mask Pixel Cloud

For the L2 HR Water Mask Pixel Cloud product, there is not one quality flag, rather each of the constituents that comprise the pixel cloud data (e.g. sigma0, interferogram, and geolocation) has its own quality flag that follows the good, suspect, degraded, and bad classification. Here, any measurements classified as bad were excluded. Additionally, any measurements with a suspect phase unwrapping geolocation quality flag or classified as land were excluded. The water level relative to NAVD88 was calculated using equation 4:

$$WSE_{NAVD88} = H - Geoid - SET - LT - PT + EGM2NAVD88 \quad (4)$$

where H is the height of the pixel above the reference ellipsoid, Geoid is the geoid height above the reference ellipsoid as reported by this product, SET is the solid earth tide height, LT is the load tide, PT is the pole tide, and EGM2NAVD88 is a local conversion from EGM2008 datum to NAVD88.

2.2 In-situ Data Processing

2.2.1 Processing In-situ WSE

In-situ WSE was measured with a microwave radar water level sensor mounted at the end of a 560 m long pier in 7 m water depth (Figure 1). The water level sensor samples at 1 Hz and reports 6-minute average water level, relative to NAVD88 (NOAA, 2026).

2.2.2 Processing In-situ SWH

Four wave gauges were used to assess the accuracy of SWOT estimates in various water depths. Two wave rider buoys 16 and 4 km offshore, with nominal depths of 26 and 17 m water depth,

provided offshore and intermediate depth SWH measurements, respectively (USACE, 2026b, 2026a). Closer to shore we used the FRF 8 m array that consists of 15 pressure sensors approximately 900 m offshore in 8 m water depth (USACE, 2026c). Further onshore is a Signature 1000 approximately 500 m offshore at a depth of 7 m (USACE, 2026d). For all four instruments, SWH data are reported as hourly averages based on 1.28, 2, or 10 Hz measurements, depending on the instrument. The hourly average measurement closest in time to each SWOT measurement was used to assess the accuracy of the SWOT SWH estimates.

2.3 Analysis

2.3.1 WSE Analysis

We compared SWOT and in-situ WSE separately for the four different products. For the LR products, we compared the SWOT WSE measurement closest to the NOAA tide gauge at the end of the pier (7 m water depth). For the two HR products, we defined an elliptical region centered on the NOAA tide gauge and took the spatial mean of the SWOT WSE inside the region. The size of the averaging region was determined by finding the ellipse major and minor axis dimensions between 0 to 2000 m with the smallest error between the spatially averaged WSE and the in-situ WSE. The HR Raster 100 m and pixel cloud products were optimized separately, but both were found to have an optimal averaging ellipse measuring 500 m in the cross-shore and 1000 m in the along shore.

The primary error metric was root mean square error (RMSE). Normalized RMSE (nRMSE) is also reported, where RMSE is normalized by the range of in-situ values used for the comparison.

158 For WSE, the observed range of in-situ measurements was 1.88 m at the NOAA tide gauge, for
159 SWH, it varied by instrument ranging from 1.73 to 2.14 m.

160 **2.3.2 SWH Analysis**

161 SWOT SWH estimates were evaluated at each of the four in-situ instruments separately. For each
162 instrument, the closest SWOT SWH measurement was used for each pass. For all four wave
163 gauges, a maximum distance filter from the in-situ measurement location to the SWOT SWH
164 measurement was set at 1500 m. For passes with no SWH measurement within this distance, no
165 comparison was made. In-situ and SWOT SWH measurement were both separately filtered to
166 remove any measurements greater than three standard deviations from the mean before
167 comparing.

168 In addition to relationships between environmental factors and WSE accuracy, we also
169 investigated SWOT SWH error as a function of water depth, SWOT SWH measurement distance
170 to shore, and product version. We extended this to a random forest feature importance analysis
171 to investigate which factors were contributing to SWH accuracy degradation. We used these
172 results to implement and test several additional filtering techniques. We identified a strong link
173 between accuracy degradation and distance to coast. Consequently, to determine if there was a
174 specific point where SWOT meavisoasurements get too close to shore and become erroneous,
175 we tested how the SWOT SWH accuracy is affected when implementing a distance to coast filter.
176 To do this, we processed the SWH normally but applied increasing distance to coast filters from
177 0 to 10 km from the coast at 100 m increments, each time reselecting the SWOT measurement
178 closest to each of the four wave gauges that satisfied the distance to coast and distance to in-situ
179 gauge requirements. We then compared the SWH errors over the varying distance to coast filter

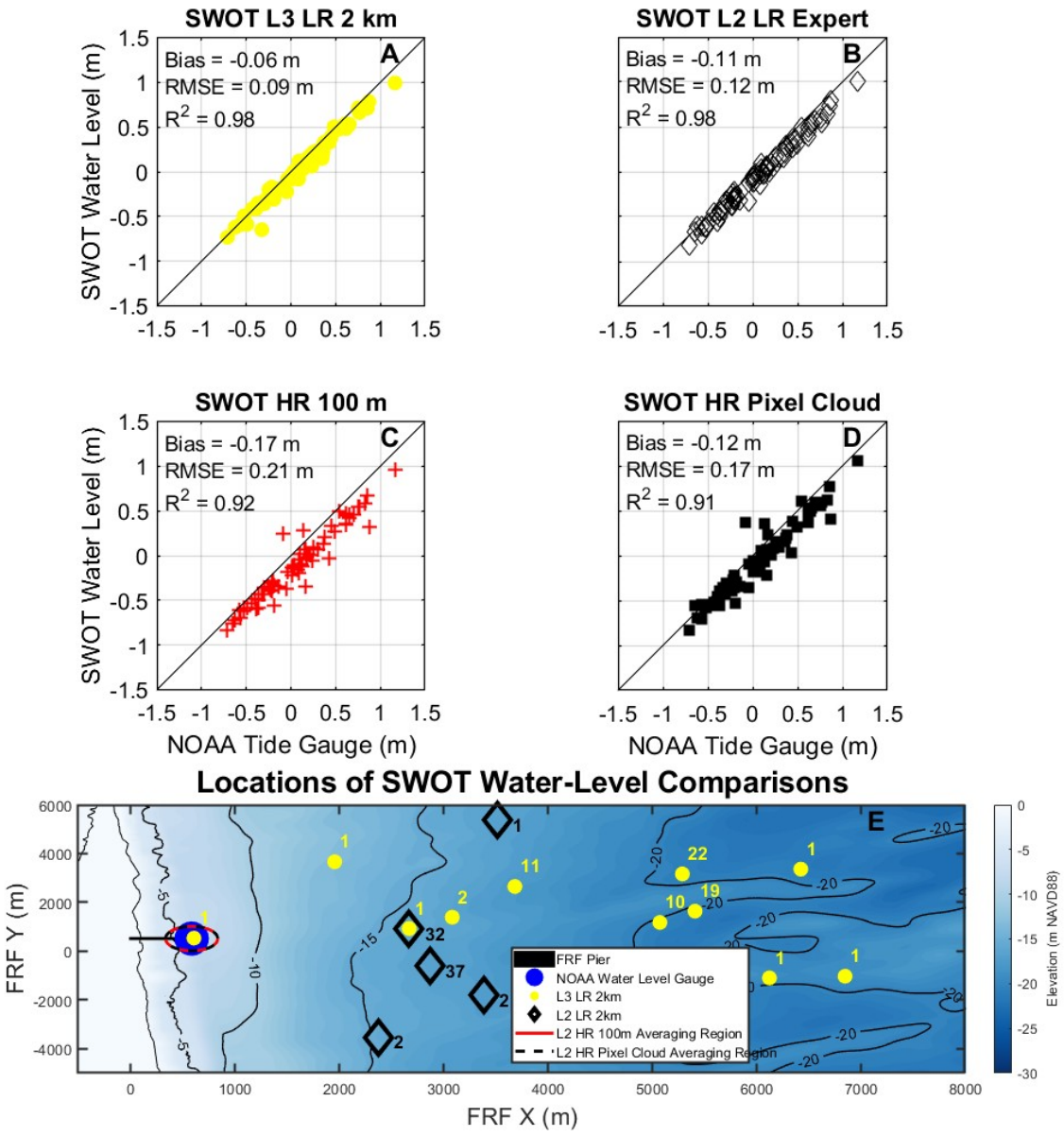
values to identify the best filter distance. We then reprocessed the SWOT L2 LR SSH Expert with the optimal distance to coast filter value applied.

3 Results and Discussion

3.1 WSE

SWOT WSE from the LR (L3 and L2) products were accurate (RMSE= 0.09 and 0.16 m, nRMSE=4.8% and 6.6%) compared to the microwave radar water level gauge (Figure 2a-d). Accuracy decreases for the HR (100 m raster and pixel cloud) products (RMSE=0.21 and 0.17 m, nRMSE=11.2% and 9.3%). The bias increased from -0.06 to -0.17 m from the L3 LR product to the HR 100 m raster product, where a negative bias indicates that the SWOT WSE was less than the in-situ measurement. We observe that the LR products are more accurate than the HR products, but the locations of the measurements are much farther offshore than the water level sensor, as compared to the HR products, where all of the SWOT comparison measurements are right at the location of the water level sensor (Figure 2e). For all products, the time differences between SWOT and in-situ measurements were maximum of three minutes and therefore the associated error contribution is expected to be negligible. We also observe that despite the HR raster 100 m product being a more refined product than the HR pixel cloud product, it has a larger error, likely due to the larger number of measurements averaged when using the pixel cloud product. Averaging the HR 100 m raster product, rather than just using a single measurement, offers only a negligible (<5%) decrease in error from 0.22 to 0. 21 m RMSE, whereas averaging the HR pixel cloud product decreases the RMSE by 0.10 m, from 0.27 to 0.17 m, a 37% reduction.

201 Other studies report L3 LR SSH Expert WSE errors of 0.14 in the Bristol Channel and Severn
202 River-Estuary (Lichtman et al., 2025) and 0.11 m on the US West Coast in the coastal zone, 0-
203 5km from shore (Zhu et al., 2025). In the English Channel in 8-12 m water depth, 3-4 km
204



205

206 *Figure 2. SWOT WSE (y axis) vs. NOAA Tide Gauge Water Level Measurement (x axis) for (a) SWOT*
207 *Level 3 LR SSH Expert (2 km), (b) SWOT Level 2 LR SSH Expert (2 km), (c) SWOT Level 2 HR 100m*
208 *Raster, and (d) SWOT Level 2 HR Pixel Cloud and the e) locations of SWOT measurements used for*

comparison across the four different products labeled with the number of times each location was used

offshore Turki et al. (2025) report WSE errors of 0.9 m using the L2 LR SSH Expert product. Rong et. al. (2026) report a WSE RMSE of 0.4 m when comparing SWOT pixel cloud WSE measurements to airborne LiDAR WSE measurements averaged over a 40 m radius in the Bristol Channel and Severn Estuary, which has a much larger tidal range compared to the FRF in Duck, NC, but with similar wave climates.

Accuracy of WSE measurements less than 500 m offshore (7 m water depth) across the HR 100 m raster and pixel cloud products suggest that they are usable in coastal environments (e.g., RMSE= 0.21 and 0.17 m, nRMSE=11% and 9.3%). The LR products offer better absolute accuracy (RMSE=0.09 and 0.12 m, nRMSE=4.8% and 6.6%) but are 3-5 km offshore. These results suggest that if absolute accuracy is the foremost goal, the L3 LR SSH Expert product should be used (RMSE=0.09 m, nRMSE=4.8%). But for use in complex topographic environments such as coastlines, the HR products still offers reasonably accurate WSE measurements (RMSE = 0.17 m, nRMSE=9.3%). Specifically, the HR 100 m raster product has a footprint of 100 by 100 m, with a RMSE of 0.22 m (nRMSE=11.7%) using every (n=70) valid measurement within 22 m of the in-situ sensor, which is located 500 m offshore. While this resolution is sufficient for open coastlines, even more complex areas may see a decrease in useable data and consequently WSE accuracy. For these more confined coastal cases such as deltas, estuaries, or coastal river mouths where areas of water are often smaller than 100 m, the HR pixel cloud with no spatial averaging is also a reliable source for WSE data (RMSE = 0.27 m, nRMSE=14.4% in 7 m water depth). One of the key advantages of HR pixel cloud data is that

232 there are more measurements that are likely to be valid very close to land. Here, the farthest
233 valid measurement is only 63 m away, and on average it is less than 50 m away. The accuracy of
234 SWOT WSE measurements did not exhibit any relationships to any of the environmental
235 conditions tested here such as wave, current, and wind conditions.

236 **3.2 SWH**

237 SWOT SWH from the L2 LR SSH Expert products are generally accurate (e.g., RMSE=0.19 and 0.18
238 m, nRMSE=8.9% and 8.4%) compared to in-situ measurements at the two farthest offshore buoys
239 (Figure 3a and 3b). Accuracy decreases quickly at the two onshore wave gauges (RMSE= 2.01 and
240 1.96 m, nRMSE=116% and 100%, Figure 3.3b and 3.3c). It is apparent that SWOT SWH measures
241 too close to land become erroneously large. Of the variables tested in the feature importance
242 analysis, the distance of the SWOT measurement to coastline and to the in-situ comparison had
243 the strongest influence on the SWOT SWH error. The distance from the SWOT to the in-situ
244 measurement is expected to impact accuracy because waves shoal and steepen as they approach
245 the shoreline.

246 With these insights, we redid the SWH accuracy analysis, this time imposing the 1700 m distance
247 to coast filter. For the wave gauges in 7 and 8 m water depth, this moves the cross-shore location
248 of SWOT estimates approximately 2000 m offshore of the in-situ measurement locations. To
249 account for this large cross-shore distance between the 7 and 8 m wave gauges and the closest
250 SWOT SWH once the 1700 m distance to coast filter was applied, we shoaled wave heights from
251 the SWOT location to the in-situ gauge locations (Figure 4c-d) using linear wave theory (Equation
252 5):

253 $$H_2 = H_1 \left(\frac{c_{g_1}}{c_{g_2}}\right)^{0.5} \quad (5)$$

where H_1 and H_2 are the SWOT SWH measured offshore (near the 17 m depth gauge) and shoaled onshore to the target location of the 7 and 8 m wave gauges, respectively, and $\boldsymbol{c_{g_1}}$ and $\boldsymbol{c_{g_2}}$ are the group velocities at the offshore SWOT SWH location and the target location at the 7 and 8 m wave gauges, respectively. With the 1700 m distance to coast filter, the H_1

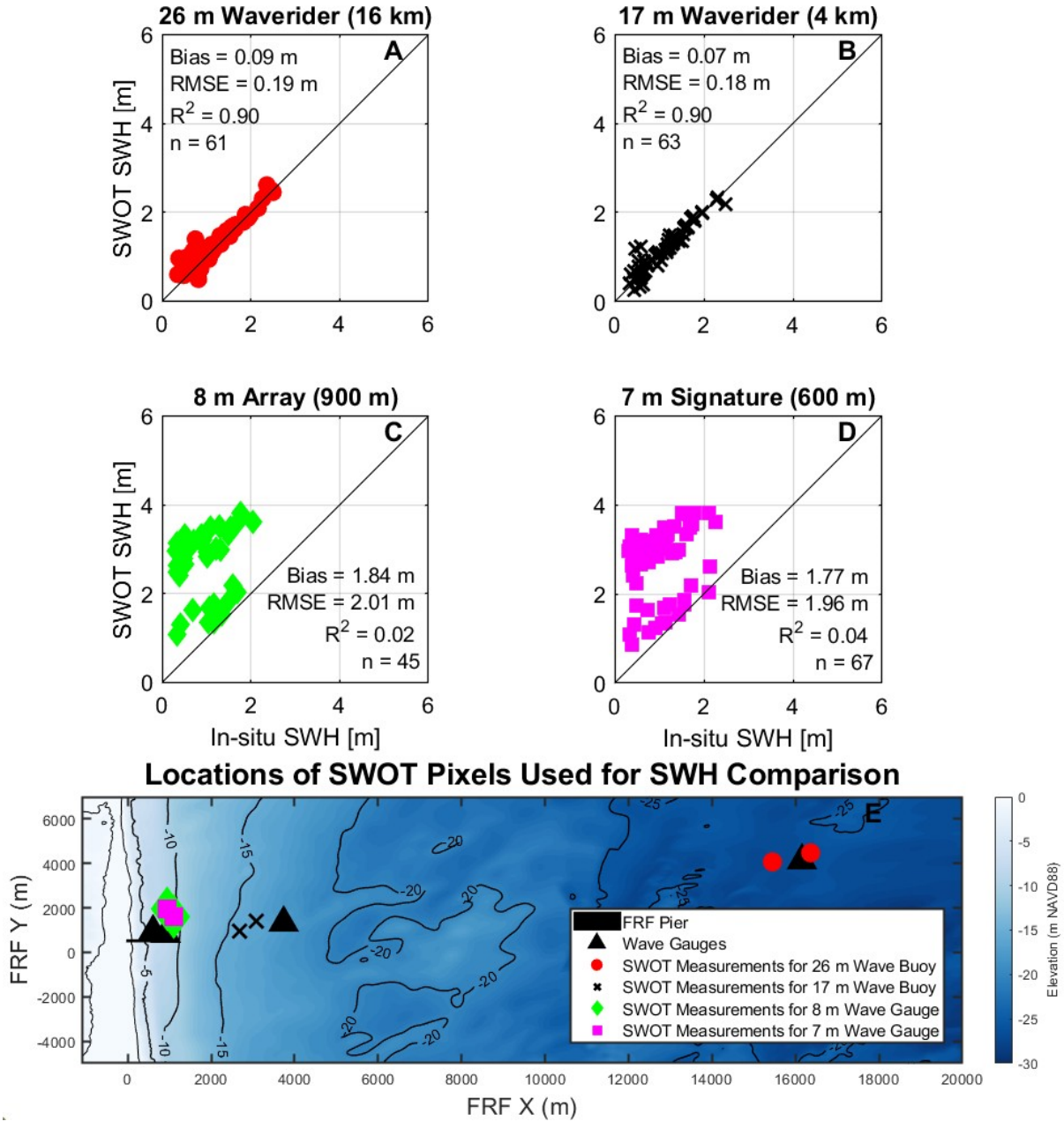


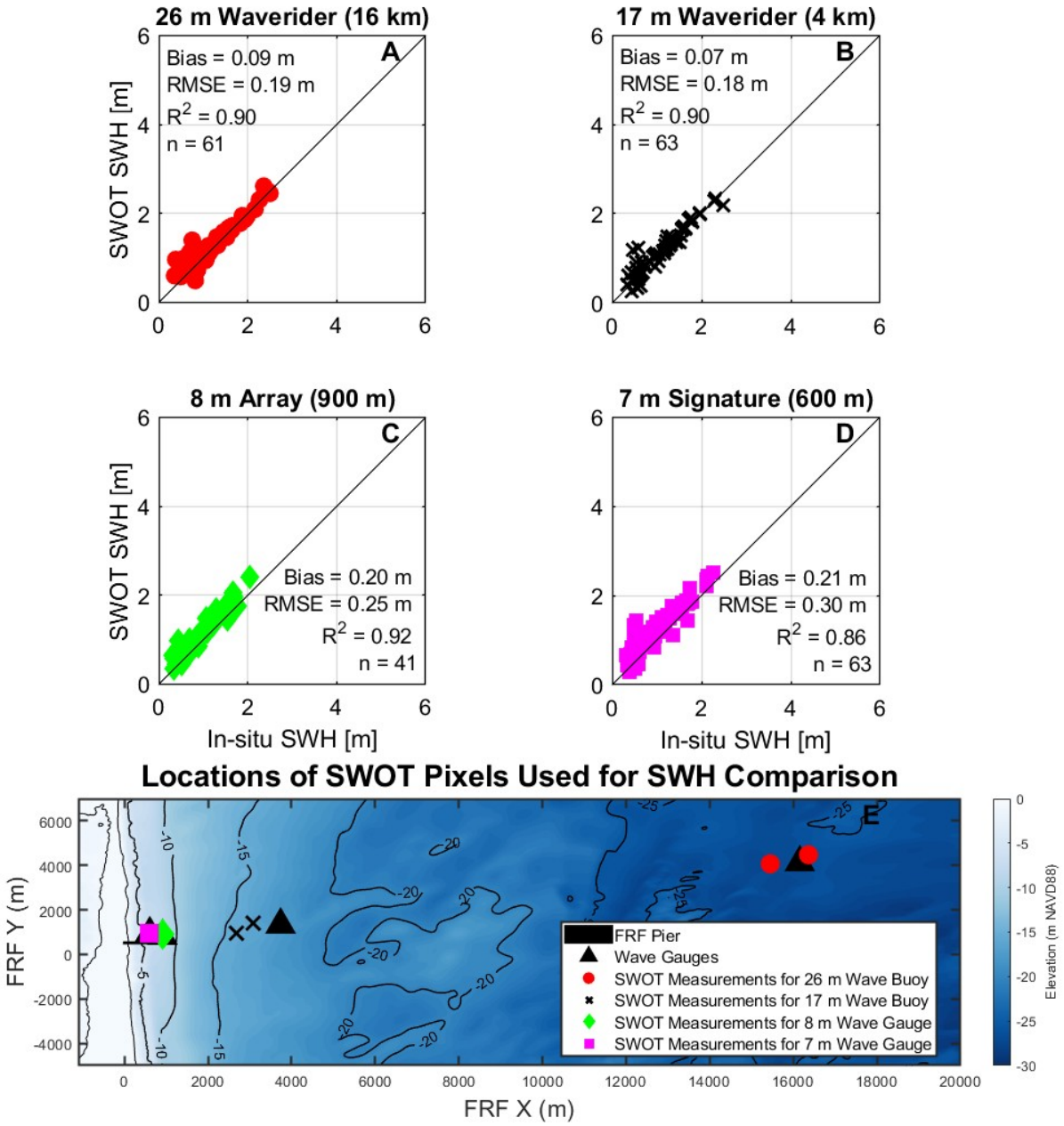
Figure 3. SWOT Significant wave height (SWH) estimates (y axis) vs. in-situ SWH measurements (x axis) at (a) Waverider buoy in 26 m water depth (16 km offshore), (b) waverider buoy in 17 m

water depth (4 km offshore), (c) 8 m wave gauge (900 m offshore), (d) 7 m wave gauge, and (e) shows the location of the SWOT measurements used for comparison for each of the four instruments.

measurements were approximately 2000 m offshore of the 7 and 8 m wave gauges before shoaling was applied. The distance to coast filter for the SWOT SWH had no effect on error at the 26 and 17 m wave buoys (Figure 4a-b) because they are much further offshore (16 and 4 km) than the measurements that are affected by land. For the 7 and 8 m wave gauges, we observe a 2 m median reduction in absolute error when limiting SWOT SWH to at least 1700 m from the coast (Figure 4c-d) because we are eliminating SWH that are contaminated by land that causes erroneously large SWH estimates. By shoaling these wave heights, it offers a more physically realistic estimate of the SWH, compared to a measurement 2000 m offshore, in a region where a few hundred meters can have significant differences in wave behavior and demonstrates one approach to estimating SWH close to shore, further extending the capabilities of SWOT in coastal environments. The results are consistent with SWOT SWH errors observed in other studies (Dong et al., 2025; Ho et al., 2025; Turki et al., 2025; Villas Bôas et al., 2025; Xie et al., 2026).

These results demonstrate that similar SWH accuracy shown for SWOT in offshore settings can be obtained in coastal environments but extra care must be taken to remove the erroneous data very close to shore (approximately1700 m). In this case, the erroneous data were not captured by quality flags but were obviously invalid evidenced by the magnitude of the SWOT SWH being three to four times the magnitude of adjacent SWOT and in-situ measurements. There may be several issues contributing to the degradation of SWOT SWH very close to shore. The primary source of SWH error is the contamination of KaRIN measurements from land pixels, as the KaRIN radar waves reflect off land surfaces away from the KaRIN sensor, compared to radar

285 waves reflecting off water surfaces towards the KaRIN sensor (Durand et. al., 2020). This
286 introduces apparent unaccounted for noise that results in larger estimated SWH with the
287 volumetric decorrelation method used by SWOT. Another source of SWH error that is particularly
288 relevant in the nearshore is wave bunching (Fernandez, 2017), where the nonuniform velocity of



289

290 *Figure 4. SWOT Significant wave height (SWH) estimates (y axis) vs. in-situ SWH measurements*
291 *(x axis) with a 1700 m distance to coast filter and linear shoaling applied to SWOT SWH*

measurements for the 8 and 7 m sensor comparisons. (a) Waverider buoy in 26 m water depth (16 km offshore), (b) waverider buoy in 17 m water depth (4 km offshore), (c) multi-sensor wave array in 8 m water depth (900 m offshore), (d) Signature 500 in 7 m water depth, and (e) shows the location of the SWOT measurements used for comparison for each of the four instruments with the 1700 m distance to coast filter applied and shoaling the wave heights to the location of 7 and 8 m wave gauges.

the waves, specifically wave bores, inflict a nonlinear shift on the apparent SSH. Additionally, as waves get steeper in the nearshore, the surfboard effect (Peral and Chen, 2021; Fernandez, 2017) becomes more prevalent, leading to measurements that are biased low due to the suppression of small scale SSH variances. Temporal differences between the SWOT and in-situ measurements may also contribute to the comparison error. The in-situ SWH measurements were averaged over 60 minutes but only SWOT passes with an in-situ measurement within 30 minutes were used. Another expected temporal error contribution is when the SWOT flyover captures a wave field that is not characteristic of the 60-minute averaging period.

4 Conclusion

We combined SWOT with in-situ data to better understand the capabilities and deficiencies of SWOT in coastal environments. We evaluated the performance of four SWOT WSE products at 500 m and one SWH product at four locations ranging from 500 m to 16 km offshore. Across the four products, WSE accuracy decreased as the level of processing increased and spatial resolution decreased with the LR L3 SSH Expert product with the lowest RMSE of 0.09 m (nRMSE=4.8%) and the L2 HR Water Mask 100 m Raster Product with the highest RMSE of 0.21 m (nRMSE=11.2%). SWH agreement was very good between SWOT and the 17 and 26 m wave buoys with RMSE of 0.19 (nRMSE=8.9%). Agreement between SWOT and the onshore wave gauges in 7 and 8 m water depth was also good (RMSE= 0.25 m, nRMSE=14.4%) once a distance to coast threshold at 1700m

316 offshore was implemented and waves were shoaled from the location of the farthest onshore
317 valid measurements to the locations of the 7 and 8 m buoys.

318 Accurate WSE estimates within 600 m of shore suggests that SWOT is not only capable of
319 measuring WSE in open water, but can also be used along the inner shelf before waves transform
320 in the nearshore. This provides exciting opportunities to use SWOT to track sea level rise of
321 coastal communities on global scales. And though the SWOT SWH accuracy decreases rapidly
322 within a kilometer of land, the accurate SWOT WSE show promise for extending accurate SWH
323 estimates closer to shore with nearshore specific SWH algorithms.

324

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Data Availability Statement

The SWOT Level 3 Low-Rate Sea Surface Height Expert used for the WSE analysis is available on CNES's AVISO platform via https://doi.org/10.24400/527896/A01-2024.003 with an AVISO+ license agreement (AVISO/DUACS, 2024). The SWOT Level 2 KaRIN Low-Rate Sea Surface Height Expert Data Product, Version C and Version D, used for the WSE and SWH analyses are available on NASA's Earth Data Platform via https://doi.org/10.5067/SWOT-SSH-2.0 and https://doi.org/10.5067/SWOT-SSH-D without restriction (SWOT, 2024a, 2025a). The Level 2 Water Mask Raster Image 100 m Data Product, Version C and Version D, used for the WSE analysis are available on NASA's Earth Data Platform via https://doi.org/10.5067/SWOT-RASTER-2.0 and https://doi.org/10.5067/SWOT-RASTER-D with restriction (SWOT, 2024c, 2025c). The SWOT Level 2 Water Mask Pixel Cloud Data Product, Version C and Version D, used for the WSE analysis are available on NASA's Earth Data Platform via https://doi.org/10.5067/SWOT-PIXC-2.0 and https://doi.org/10.5067/SWOT-PIXC-D without restriction (SWOT, 2024b, 2025b). In-situ wave height measurement data used for the SWH analysis is provided by the US Army Engineer Research and Development Center's Coastal Hydraulics Laboratory, available at chldata.erdc.dren.mil, accessed via the CHL Data THREDDS Server in the "FRF data" folder (USACE, 2026a, 2026b, 2026c, 2026d) for public use.

In-situ water level data used for the WSE analysis is available at the NOAA Tides and Currents portal via https://tidesandcurrents.noaa.gov/stationhome.html?id=8651370 (NOAA, 2026). The MATLAB code needed to reproduce this analysis is preserved at https://doi.org/10.5281/zenodo.23102497 and openly available on GitHub at https://github.com/tylermccormack-research/Applications-Limitations-and-Insights-for-SWOT-in-Wave-Dominated-Nearshore-Environments.git under a MIT License (McCormack, 2026).

Conflict of Interest Disclosure

The authors declare there are no conflicts of interest for this manuscript.

References

Ardhuin, F., Molero, B., Bohé, A., Nouguier, F., Collard, F., Houghton, I., Hay, A., Legresy, B. (2024). Phase-Resolved Swells Across Ocean Basins in SWOT Altimetry Data: Revealing Centimeter-Scale Wave Heights Including Coastal Reflection. Geophysical Research Letters 51, e2024GL109658. https://doi.org/10.1029/2024GL109658

Ardhuin, F., Postec, T., Accensi, M., Piolle, J.-F., Dodet, G., Passaro, M., De Carlo, M., Husson, R., Guitton, G., Collard, F. (2025). Sizing the largest ocean waves using the SWOT mission. Proceedings of the National Academy of Sciences 122, e2513381122. https://doi.org/10.1073/pnas.2513381122

AVISO/DUACS. (2024). SWOT Level-3 KaRIn Low Rate SSH Unsmoothed (v2.0.1) [Dataset]. CNES. https://doi.org/10.24400/527896/A01-2024.003

Bohé, A., Chen, A., Chen, C., Dubois, P., Fore, A., Molero, B., Peral, E., Raynal, M., Stiles, B., Ardhuin, F., Hay, A., Legresy, B., Lenain, L., Villas Bôas, A.B. (2025). Measuring Significant Wave Height Fields in Two Dimensions at Kilometric Scales With SWOT. IEEE Transactions on Geoscience and Remote Sensing 63, 1–19. https://doi.org/10.1109/TGRS.2025.3551605

Dooley, C., Elgar, S., Raubenheimer, B., Gorrell, L. (2025). Estimating surfzone currents with near -field optical remote sensing. Journal of Atmospheric and Oceanic Technology 42, 33-46. https://doi.org/10.1175/JTECH-D-24-0015.1

Dooley, C., Elgar, S., Raubenheimer, B. (2024). Field Observations of Surfzone Vorticity. Geophysical Research Letters 51, e2024GL111402. https://doi.org/10.1029/2024GL111402

Fernandez, D.E. (2017). *SWOT mission performance and error budget*, JPL D-79084, Revision A. Jet Propulsion Laboratory, California Institute of Technology.

Fu, L.-L., Pavelsky, T., Cretaux, J.-F., Morrow, R., Farrar, J.T., Vaze, P., Sengenes, P., Vinogradova-Shiffer, N., Sylvestre-Baron, A., Picot, N., Dibarboure, G. (2024). The Surface Water and Ocean Topography Mission: A Breakthrough in Radar Remote Sensing of the Ocean and Land Surface Water. Geophysical Research Letters 51, e2023GL107652. https://doi.org/10.1029/2023GL107652

Ho, A., Wang, J., Haines, B., Wu, A., Stalin, S. (2025). In Situ Validation of Small-Scale Spatial Variability in Significant Wave Height Observations From SWOT. Earth and Space Science 12, e2025EA004286. https://doi.org/10.1029/2025EA004286

JPL D-56411, Revision C, "SWOT Product Description Document: Level 2 KaRIn High Rate Water Mask Pixel Cloud (L2_HR_PIXC) Data Product," Jet Propulsion Laboratory Internal Document. (2025).

Laignel, B., Ayoub, N., Birol, F., Brown, S., Chao, Y., Cornuelle, B., Costa, S., De Mey, P., Estournel, C., Feddersen, F. and Giddings, S. (2015). Issues and SWOT contribution in the coastal zones and estuaries White Paper. *SWOT White Papers*.

Lichtman, I.D., Bell, P.S., Gommenginger, C., Banks, C., Calafat, F.M., Brown, J., Williams, S.D.P. (2025). Evaluating Water Levels From the Surface Water and Ocean Topography (SWOT) Mission in a Hyper-Tidal Coastal and Estuarine Environment. Earth and Space Science 12, e2024EA004104. https://doi.org/10.1029/2024EA004104

McCormack, T. (2026). tylermccormack-research/Applications-Limitations-and-Insights-for-SWOT-in-Wave-Dominated-Nearshore-Environments: Zenodo Archive (Version v1.0.0) [Computer software]. Zenodo. https://doi.org/10.5281/zenodo.23102497

Morrow, R., Fu, L.-L., Ardhuin, F., Benkiran, M., Chapron, B., Cosme, E., d'Ovidio, F., Farrar, J.T., Gille, S.T., Lapeyre, G., Le Traon, P.-Y., Pascual, A., Ponte, A., Qiu, B., Rascle, N., Ubelmann, C., Wang, J., Zaron, E.D., (2019). Global Observations of Fine-Scale Ocean Surface Topography With the Surface Water and Ocean Topography (SWOT) Mission. Front. Mar. Sci. 6. https://doi.org/10.3389/fmars.2019.00232

NOAA. (2026). Verified 6-minute water level data for Duck, NC (station 8651370) [Dataset]. NOAA Tides and Currents, Center for Operational Oceanographic Products and Services. https://tidesandcurrents.noaa.gov/stationhome.html?id=8651370

Peral, E., Chen, C. (2021). Surface Wave Error on SWOT - global and regional impacts.

Raubenheimer, B., Guza, R.T., Elgar, S., 1996. Wave transformation across the inner surf zone. Journal of Geophysical Research: Oceans 101, 25589–25597. https://doi.org/10.1029/96JC02433

Rong, Y., Bates, P., Neal, J., Bell, P., Gommenginger, C., Lichtman, I.D., Banks, C., Calafat, F.J.M., Martin, A., Brown, J., Cooke, R., Balfour, C., De Dominicis, M., Payo Payo, M., Williams, S., Amoudry, L., Chuter, S., Neill, S., Chisholm, J., Gleason, C.J., Kupiec, J., Lewis, M. (2026). Evaluating SWOT in the Coastal Zone: Comparisons With Tide Gauge and Airborne LiDAR in the Bristol Channel and Severn Estuary, UK. Geophysical Research Letters 53, e2025GL116590. https://doi.org/10.1029/2025GL116590

SWOT. (2024a). SWOT Level 2 KaRIn Low Rate Sea Surface Height Data Product Version C [Dataset]. NASA Physical Oceanography Distributed Active Archive Center. https://doi.org/10.5067/SWOT-SSH-2.0

SWOT. (2024b). SWOT Level 2 Water Mask Pixel Cloud Data Product Version C [Dataset]. NASA Physical Oceanography Distributed Active Archive Center. https://doi.org/10.5067/SWOT-PIXC-2.0

SWOT. (2024c). SWOT Level 2 Water Mask Raster Image Data Product Version C [Dataset]. NASA Physical Oceanography Distributed Active Archive Center. https://doi.org/10.5067/SWOT-RASTER-2.0

SWOT. (2025a). SWOT Level 2 KaRIn Low Rate Sea Surface Height Data Product Version D [Dataset]. NASA Physical Oceanography Distributed Active Archive Center. https://doi.org/10.5067/SWOT-SSH-D

SWOT. (2025b). SWOT Level 2 Water Mask Pixel Cloud Data Product Version D [Dataset]. NASA Physical Oceanography Distributed Active Archive Center. https://doi.org/10.5067/SWOT-PIXC-D

SWOT. (2025c). SWOT Level 2 Water Mask Raster Image Data Product Version D [Dataset]. NASA Physical Oceanography Distributed Active Archive Center. https://doi.org/10.5067/SWOT-RASTER-D

Turki, E.I., Islam, M.S., Solano, C.L., Salameh, E., Domingues, M., Mendoza, T., Aouf, L., Froideval, L., Barcelo, D.G., Laignel, B., Carbonniere, A., Frappart, F. (2025). On the use of SWOT altimetry data for monitoring coastal hydrodynamics. International Journal of Applied

Earth Observation and Geoinformation 145, 104982. https://doi.org/10.1016/j.jag.2025.104982

USACE. (2026a). FRF 17m Datawell Waverider Buoy [Dataset]. United States Army Corps of Engineers Coastal and Hydraulics Laboratory Data THREDDS Server. https://chldata.erdc.dren.mil/thredds/catalog/frf/oceanography/waves/waverider-17m/catalog.html

USACE. (2026b). FRF 26m Datawell Waverider Buoy [Dataset]. United States Army Corps of Engineers Coastal and Hydraulics Laboratory Data THREDDS Server. https://chldata.erdc.dren.mil/thredds/catalog/frf/oceanography/waves/waverider-26m/catalog.html

USACE. (2026c). FRF 8m Array [Dataset]. United States Army Corps of Engineers Coastal and Hydraulics Laboratory Data THREDDS Server. https://chldata.erdc.dren.mil/thredds/catalog/frf/oceanography/waves/8m-array/catalog.html

USACE. (2026d). FRF Signature940-600 [Dataset]. United States Army Corps of Engineers Coastal and Hydraulics Laboratory Data THREDDS Server. https://chldata.erdc.dren.mil/thredds/catalog/frf/oceanography/waves/sig940-600/catalog.html

Wang, J., Lucas, A.J., Stalin, S., Lankhorst, M., Send, U., Schofield, O., Kachelein, L., Haines, B., Meinig, C., Pinkel, R., Farrar, J.T., Fu, L.-L. (2025). SWOT Mission Validation of Sea Surface Height Measurements at Sub-100 km Scales. Geophysical Research Letters 52, e2025GL114936. https://doi.org/10.1029/2025GL114936

462 Wang, Y., Zhang, S., Jia, Y. (2025). Enhanced resolution capability of SWOT sea surface height
463 measurements and their application in monitoring ocean dynamics variability. Ocean
464 Science 21, 931–944. https://doi.org/10.5194/os-21-931-2025
465 Xie, H., Li, X., Jiao, W., Liu, B., & Lang, S. (2026). An integrated evaluation for SWOT KaRIn Level-
466 2 significant wave height product. GIScience & Remote Sensing, 63(1), 2614566.
467 https://doi.org/10.1080/15481603.2026.2614566
468 Zhu, H., Peng, F., & Shen, Y. (2025). Validation of Sea Level Anomalies from the SWOT
469 Altimetry Mission Around the Coastal Regions of East Asia and the US West Coast.
470 *Water*, *17*(21). https://doi.org/10.3390/w17213066
471